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A REPORT ON

**“Design and Implementation of a Machine Learning Based System for
Early Detection and Accurate Classification of Breast Cancer Using
Diagnostic Data”**

Submitted in the partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY IN
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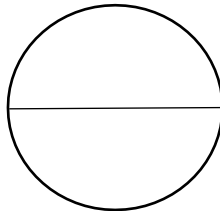
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CERTIFICATE

This is to certify that, **Mr. Andanappa subhasha santoji (01SU24AI013)** has satisfactorily completed the assessment (Report) in **ADVANCED MACHINE LEARNING (25SAL051)** prescribed by the Srinivas University for the 5th semester B. Tech course during the year **2026**.

MARKS AWARDED



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1. INTRODUCTION

Machine Learning (ML) is an important field of Artificial Intelligence (AI) that enables computer systems to learn useful information from existing data and use that knowledge to produce predictions or decisions. Unlike traditional programming, where specific instructions are provided for every task, machine learning algorithms can identify relationships and patterns within data automatically. ML techniques are widely applied in areas such as medical diagnosis, image processing, financial analysis, speech recognition, recommendation systems, and pattern identification. Supervised learning is one of the major categories of machine learning and is particularly useful when the training data contains known output labels.

Classification is a supervised learning approach used to place data samples into predefined categories based on their input characteristics. During the learning process, the algorithm studies labelled training examples and develops a model that can classify new and unseen observations. Different classification algorithms use different mathematical and logical approaches, which can result in variations in prediction accuracy and reliability. Therefore, testing several algorithms on the same dataset provides a better basis for evaluating their strengths and weaknesses and selecting a suitable model for the given problem.

The Breast Cancer Dataset is another important dataset considered in this project and is commonly used for binary classification and medical data analysis. It contains numerical measurements obtained from breast tissue samples, with the target variable indicating whether a tumour is classified as malignant or benign. The dataset includes multiple features describing characteristics of the cell nuclei, allowing classification algorithms to learn patterns associated with the two diagnostic categories. Using this dataset makes it possible to examine how the selected algorithms perform on structured medical data in addition to handwritten image data.

Three machine learning classification algorithms are implemented and evaluated in this project: Decision Tree, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes. These methods differ in their learning principles and classification strategies. The datasets are divided into appropriate training and testing portions so that the models can be trained using one part of the data and evaluated on previously unseen samples. Feature scaling is also applied where necessary, particularly for algorithms such as KNN that are sensitive to differences in feature magnitudes.

The performance of the selected algorithms is measured using several commonly used evaluation metrics, including Accuracy, Precision, Recall, and F1-Score. Confusion matrices are also generated to provide a detailed representation of correct and incorrect predictions for the respective classes. These evaluation measures help in understanding not only the overall performance of each model but also the specific classification errors made by the algorithms. Comparing these results across both datasets provides a broader view of algorithm performance under different types of classification problems.

The main purpose of this project is to analyse and compare the effectiveness of Decision Tree, KNN, and Gaussian Naive Bayes on the Digits and Breast Cancer datasets. The comparison helps identify which algorithm provides better classification performance for each dataset. Through this experiment, practical knowledge is gained in important stages of a machine

learning workflow, including dataset preparation, feature processing, model training, prediction, evaluation, confusion matrix analysis, and comparative performance assessment.

2. DATASET DESCRIPTION

The dataset used for this project is the Breast Cancer Wisconsin (Diagnostic) Dataset, which is a widely used dataset for machine-learning classification. The main purpose of the dataset is to classify breast tumours as either malignant or benign using numerical measurements obtained from digitized images of breast tissue samples

The dataset contains 569 instances and 30 numerical features. Each instance represents a breast tissue sample, while the features describe different characteristics of the cell nuclei.

The target variable is Diagnosis, which has two classes: M (Malignant) and B (Benign).

Malignant represents a cancerous tumour, while benign represents a non-cancerous tumour. Since there are two possible output classes, the dataset is considered a binary classification dataset.

2.1 Dataset Features

The 30 numerical features in the dataset are derived from ten basic characteristics of the cell nuclei. These characteristics are measured in three forms: mean value, standard error, and worst or largest value. The ten basic characteristics are:

- Radius: Measures the distance from the centre of the nucleus to its boundary.
- Texture: Represents the variation in grayscale intensity of the nucleus.
- Perimeter: Measures the boundary length of the nucleus.
- Area: Represents the size of the cell nucleus.
- Smoothness: Describes the variation in the radius along the boundary.
- Compactness: Represents the compactness and shape of the nucleus.
- Concavity: Measures the severity of concave portions of the nucleus boundary.
- Concave Points: Represents the number or degree of concave regions.
- Symmetry: Measures the symmetry of the cell nucleus.
- Fractal Dimension: Describes the complexity of the boundary of the nucleus.

2.2 Dataset Summary

The important properties of the dataset are shown below:

Attribute	Description
Dataset Name	Breast Cancer Wisconsin (Diagnostic) Dataset
Number of Instances	569
Number of Features	30

Feature Type	Numerical
Attribute	Description
Target Variable	Diagnosis
Number of Classes	2
Malignant Class	M
Benign Class	B
Problem Type	Binary Classification

2.3 Target Variable

The diagnosis attribute is the target variable of the dataset. It indicates whether a tumour is malignant or benign. The two classes are represented by M and B.

The target variable is separated from the input features during machine-learning model development. The 30 numerical features are provided to the model as input, and the model learns patterns that can be used to predict the diagnosis of new samples.

For machine-learning purposes, the categorical values can be converted into numerical labels. This allows classification algorithms to process the target variable efficiently. The model can then be trained using known samples and tested using previously unseen samples.

2.4 Data Preprocessing

Before using the dataset for machine learning, preprocessing is performed to improve the quality and consistency of the data. The dataset should first be checked for missing values, duplicate records, and irrelevant attributes. Any unnecessary identification column can be removed because it does not contribute to tumour classification. Feature scaling may also be performed because the numerical features have different ranges. Techniques such as standardization can be used to place the features on a comparable scale. This is particularly important for algorithms such as Support Vector Machine, K-Nearest Neighbours, and Logistic Regression.

2.5 Use of the Dataset

The Breast Cancer Dataset can be used to develop and compare different classification algorithms. Common algorithms include Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours (KNN), Support Vector Machine (SVM), and Neural Networks.

The performance of these algorithms can be evaluated using measures such as accuracy, precision, recall, F1-score, and confusion matrix. These measures help determine how effectively a model distinguishes between malignant and benign tumours.

2.6 Summary

The Breast Cancer Wisconsin (Diagnostic) Dataset is a structured dataset containing 569 samples and 30 numerical features. The features describe characteristics of breast cell nuclei, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave

points, symmetry, and fractal dimension. The target variable, Diagnosis, classifies the samples into two categories: malignant (M) and benign (B).

3. MACHINE LEARNING ALGORITHMS USED

Machine learning algorithms are used in this project to analyse the Breast Cancer Dataset and classify tumour samples into two categories: **benign and malignant**. Since the dataset contains labelled data, supervised machine learning techniques are suitable for this classification task. Three different classification algorithms are selected in this project: **Decision Tree Classifier, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes**.

The purpose of using three algorithms is to compare their ability to identify the class of a tumour based on the given numerical features. Although all three algorithms perform the same classification task, they use different methods for learning patterns from the data. The models are trained using the training dataset and then tested using unseen data. Their performance is compared using **Accuracy, Precision, Recall, and F1-Score**.

3.1 Decision Tree Classifier

The **Decision Tree Classifier** is a supervised machine learning algorithm that makes predictions by creating a sequence of decisions based on the input features. It represents the decision-making process in the form of a tree. The top of the tree is called the root node, intermediate points are decision nodes, and the final results are represented by leaf nodes.

For the Breast Cancer Dataset, the Decision Tree analyses features such as radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry. It selects suitable features and threshold values to divide the samples into groups. These divisions continue until the model reaches a final classification.

For example, if a particular feature value separates benign and malignant samples effectively, the Decision Tree can use that feature as a decision point. By following the sequence of decisions from the root to a leaf, the model determines whether a new tumour sample is benign or malignant.

One of the important advantages of a Decision Tree is that its decisions can be easily understood. Unlike some complex models, the classification process can be represented through simple rules. It also does not require feature scaling, which makes it convenient for numerical datasets.

Advantages

- Simple to understand and interpret.
- Easy to implement for classification tasks.
- Does not require feature scaling.
- Can work with different types of data.
- Provides understandable decision rules.
- Can help identify important features.

However, a Decision Tree can overfit the training data if the tree becomes unnecessarily large or complex. Therefore, parameters such as maximum tree depth and minimum samples per leaf can be used to control the complexity of the model.

3.2 K-Nearest Neighbours (KNN)

K-Nearest Neighbours (KNN) is a supervised classification algorithm that predicts the class of a new sample by examining the most similar samples in the training dataset. The similarity between samples is usually determined using a distance measure.

In this project, KNN is applied to classify breast tumour samples as benign or malignant. When a new sample is provided, the algorithm calculates its distance from the training samples and identifies the nearest neighbours. The class that occurs most frequently among these neighbours becomes the predicted class.

The value of **K** determines how many neighbouring samples are considered. For example, when $K = 5$, the algorithm considers the five closest training samples. If most of these neighbours are malignant, the new sample is classified as malignant. If most are benign, it is classified as benign.

Feature scaling is an important preprocessing step for KNN. The Breast Cancer Dataset contains features with different numerical ranges. Since KNN depends on distance calculations, features with larger values could dominate the distance measurement if scaling is not performed. Therefore, standardization can be applied before training the KNN model.

Advantages

- Simple and easy to understand.
- Easy to implement.
- Suitable for numerical datasets.
- Useful for classification problems.
- Does not require a complicated training process.
- Can classify new samples based on similarity.

The main disadvantage of KNN is that prediction can become slower as the number of training samples increases. Its performance is also influenced by the selected value of K and the distance measure used.

3.3 Gaussian Naive Bayes

Gaussian Naive Bayes is a probability-based classification algorithm derived from Bayes' theorem. It calculates the probability of a sample belonging to each possible class and selects the class with the highest probability.

For this project, Gaussian Naive Bayes is used to determine whether a breast tumour sample belongs to the benign or malignant class. The algorithm uses the numerical features of the tumour sample to calculate the probability of each class.

The Gaussian version of Naive Bayes assumes that the numerical features follow a normal or Gaussian distribution within each class. It also assumes that the features are conditionally independent of one another when the class is known. Although this assumption may not always be completely true, the algorithm can still provide effective results for many classification problems.

One of the major strengths of Gaussian Naive Bayes is its computational efficiency. It requires relatively little training time and can make predictions quickly. This makes it useful for comparing the performance of a simple probabilistic model with other classification techniques.

Advantages

- Fast and computationally efficient.
- Simple to implement.
- Suitable for numerical features.
- Requires relatively little training time.
- Can handle datasets with multiple features.
- Provides probability-based classification.

A limitation of this algorithm is that its performance depends on the assumptions about feature distributions and independence. If the features are strongly related to each other or do not follow the assumed distribution, the classification performance may be affected.

3.5 Performance Evaluation

After training, the three models are tested on the testing dataset. Their performance is measured using **Accuracy, Precision, Recall, and F1-Score**.

Accuracy represents the proportion of correctly classified samples out of all tested samples.

Precision indicates how accurately the model identifies samples predicted as a particular class.

Recall measures the ability of the model to correctly identify samples belonging to the required class. The **F1-Score** combines precision and recall into a single measure.

A **confusion matrix** can also be used to understand the correct and incorrect predictions made by each algorithm. Comparing these results provides a better understanding of the strengths and weaknesses of the selected models.

3.6 Conclusion

The three supervised machine learning algorithms used in this project provide different methods for classifying breast tumours. The Decision Tree focuses on decision rules, KNN uses similarity between samples, and Gaussian Naive Bayes uses probability.

Applying these algorithms to the same Breast Cancer Dataset makes it possible to compare their classification performance fairly. The comparison based on Accuracy, Precision, Recall, F1-Score, and the confusion matrix help identify the most suitable algorithm for the given

dataset. This approach also demonstrates how different machine-learning techniques can be applied to a real-world binary classification problem.

4. DATA PREPROCESSING

Data preprocessing is an important step in machine learning that prepares the dataset for model training and improves the reliability of predictions. In this project, the **Breast Cancer Wisconsin (Diagnostic) Dataset** is prepared before applying the selected classification algorithms. The dataset contains numerical measurements of breast cell nuclei, with the target variable indicating whether a tumor is **benign or malignant**.

First, the dataset is loaded and examined to understand its structure. The input features (**X**) and target variable (**y**) are then separated. The dataset contains 30 numerical features, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. The target variable contains two classes: **Benign (B)** and **Malignant (M)**.

The dataset is checked for missing values, duplicate records, and unnecessary columns. If an identification column is present, it is removed because it does not provide useful information for tumour classification. This step ensures that only relevant information is provided to the machine learning models. The prepared dataset is divided into **80% training data and 20% testing data** using the `train_test_split()` function. The training data is used to teach the models to identify patterns between the tumour features and their corresponding classes. The testing data is kept separate and is used to evaluate the performance of the trained models on unseen samples.

Feature scaling is performed using **StandardScaler**. The features in the dataset have different numerical ranges, so standardization converts them to a common scale. This step is particularly important for **K-Nearest Neighbour (KNN)** because KNN uses distance calculations to identify nearby samples. Scaling prevents features with larger numerical values from having an excessive influence on the distance calculation. The standardized features are also used for Gaussian Naive Bayes in this project.

The **Decision Tree Classifier** does not require feature scaling because it makes decisions using feature-based splitting rather than distance calculations. Therefore, the Decision Tree can be trained using the original feature values, while KNN uses the standardized values.

Preprocessing Steps

1. Load the Breast Cancer Dataset.
2. Examine the dataset and remove unnecessary columns.
3. Check for missing and duplicate values.
4. Separate the input features (**X**) and target variable (**y**).
5. Split the dataset into 80% training and 20% testing data.
6. Standardize the required features using **StandardScaler**.
7. Use the prepared data for training and testing the three classification algorithms.

Proper preprocessing ensures that the dataset is in a suitable form for machine learning. It helps the selected algorithms—**Decision Tree, KNN, and Gaussian Naive Bayes**—process the data effectively and provides a reliable basis for comparing their classification performance.

5. Code

```
# =====  
# ML ALGORITHM COMPARISON  
# Dataset:  
# Breast Cancer Dataset #  
Algorithms:  
# Decision Tree, KNN, Gaussian Naive Bayes  
# =====  
  
# Importing libraries  
  
import pandas as pd import matplotlib.pyplot as plt from sklearn.datasets import  
load_breast_cancer from sklearn.model_selection import train_test_split from  
sklearn.preprocessing import StandardScaler from sklearn.tree import  
DecisionTreeClassifier from sklearn.neighbors import KNeighborsClassifier from  
sklearn.naive_bayes import GaussianNB from sklearn.metrics import  
accuracy_score, precision_score, recall_score from sklearn.metrics import  
f1_score, confusion_matrix, ConfusionMatrixDisplay  
# =====  
# Load Dataset  
# =====  
# Load the Breast Cancer dataset  
data = load_breast_cancer() #  
Store features and target values X  
= data.data y = data.target  
# Display dataset information  
print("Dataset Shape:", X.shape)  
print("Classes:", data.target_names)  
# =====  
# Split Dataset  
# =====  
# Split dataset into training and testing data  
# 80% is used for training and 20% for testing  
X_train, X_test, y_train, y_test = train_test_split(  
X, y, test_size=0.20, random_state=42,  
stratify=y)
```

```

# =====

# =====

# Scaling

# StandardScaler converts features to a common scale scaler
= StandardScaler()
# Fit scaler on training data and transform it
X_train_scaled = scaler.fit_transform(X_train)
# Transform test data using the same scaler
X_test_scaled = scaler.transform(X_test)
# =====

# Create Models
# =====

# Create three machine learning classification models models
= {
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Gaussian Naive Bayes": GaussianNB()
}
# =====

# Train and Evaluate
# =====
results = []
# Train and test each algorithm for
name, model in models.items():
    # Decision Tree does not require feature scaling
    if name == "Decision Tree":
        model.fit(X_train, y_train)
        prediction = model.predict(X_test)
    # KNN and Naive Bayes
    use scaled data else:
        model.fit(X_train_scaled, y_train)
        prediction = model.predict(X_test_scaled)
    # Calculate performance metrics
    results.append([
        name,
        accuracy_score(y_test, prediction),
        precision_score(y_test, prediction),
        recall_score(y_test, prediction),
        f1_score(y_test, prediction)
    ])

# Display Results

# Convert results into a DataFrame results
= pd.DataFrame(

```

```

# =====

# =====

    results,
columns=[
    "Algorithm",
    "Accuracy",
    "Precision",
    "Recall",
    "F1-Score"
]
)
# Display algorithm comparison
print("\n===== ALGORITHM COMPARISON =====")
print(results.round(4).to_string(index=False))
# =====
# Best Algorithm
# =====
# Find the algorithm with the highest F1-Score best
= results.loc[results["F1-Score"].idxmax()]
print("\n===== BEST ALGORITHM =====")
print("Algorithm:", best["Algorithm"])    print("F1-
Score:", round(best["F1-Score"], 4))
# =====
# Performance Graph
# =====
# Create a bar graph comparing all algorithms
results.set_index("Algorithm").plot(
kind="bar",  figsize=(8, 5)
)
plt.title("ML Algorithm Performance")
plt.ylabel("Score") plt.ylim(0, 1.05)
plt.xticks(rotation=0)
plt.grid(axis="y",      alpha=0.3)
plt.tight_layout( plt.show()

| Page

# Confusion Matrix Graphs

# Generate confusion matrix for each algorithm for
name, model in models.items():
    # Train and predict using the appropriate data
    if name == "Decision Tree":

```

```

# =====

# =====

model.fit(X_train, y_train)      prediction =
model.predict(X_test)   else:
    model.fit(X_train_scaled,      y_train)
prediction = model.predict(X_test_scaled)
    # Calculate confusion matrix   cm =
confusion_matrix(y_test, prediction)   #
Display confusion matrix as a graph   disp
= ConfusionMatrixDisplay(
confusion_matrix=cm,
display_labels=data.target_names
)
    disp.plot()
    # Add title to the graph
plt.title(name + " - Confusion Matrix")
plt.tight_layout()   plt.show()

```

6. EVALUATION METRICS

Evaluation metrics are used to determine how effectively the machine learning models classify breast tumour samples. In this project, the performance of Decision Tree, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes is evaluated using Accuracy, Precision, Recall, and F1-Score. **Accuracy:** Accuracy represents the overall correctness of the classification model. It is calculated by dividing the number of correctly predicted samples by the total number of samples. A higher accuracy means that the model has correctly classified a larger proportion of the test samples.

Formula: $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

Here, TP represents True Positive, TN represents True Negative, FP represents False Positive, and FN represents False Negative.

6.1 Precision : Precision indicates how reliable the positive predictions made by the model are. It measures the proportion of correctly predicted positive samples among all samples predicted as positive. Higher precision means that the model produces fewer false-positive predictions.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

For this project, precision helps evaluate how accurately the model identifies samples belonging to the selected tumour class.

6.2 Recall : Recall, also called Sensitivity, measures the ability of the model to correctly identify actual positive samples. A high recall indicates that the model is able to detect most of the samples belonging to the positive class.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Recall is an important metric in breast cancer classification because missing a malignant tumour can be more serious than incorrectly classifying a benign tumour as malignant.

6.3 F1-Score : The F1-Score combines Precision and Recall into a single performance measure. It is useful when both precision and recall need to be considered together. A higher F1-Score indicates a better balance between the two measures.

Formula:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

6.4 Confusion Matrix

A Confusion Matrix is also used to examine the classification results of each algorithm. It shows the number of correct and incorrect predictions for the benign and malignant classes.

The matrix helps identify errors such as false positives and false negatives.

The four main outcomes are:

- True Positive (TP): The sample is correctly classified as positive.
- True Negative (TN): The sample is correctly classified as negative.
- False Positive (FP): A negative sample is incorrectly classified as positive.
- False Negative (FN): A positive sample is incorrectly classified as negative.

6.5 Comparison of Models

The values of Accuracy, Precision, Recall, and F1-Score are calculated for the Decision Tree, KNN, and Gaussian Naive Bayes algorithms using the test dataset. The results are then compared to identify the algorithm that provides the most effective classification. A model with high values across these evaluation metrics is considered to have better overall performance. However, for breast cancer classification, Recall and the number of false-negative predictions should also be carefully considered along with overall accuracy. Thus, these evaluation metrics provide a comprehensive way to assess and compare the three machine learning algorithms used in this project.

7. RESULTS AND PERFORMANCE COMPARISON

After preprocessing the Breast Cancer Dataset, three machine learning algorithms—Decision Tree, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes—were trained and tested. The performance of the models was evaluated using Accuracy, Precision, Recall, and F1-Score.

The results obtained from the test dataset are compared to determine which algorithm provides the best classification performance. The comparison helps identify the model that can most effectively distinguish between benign and malignant tumour samples.

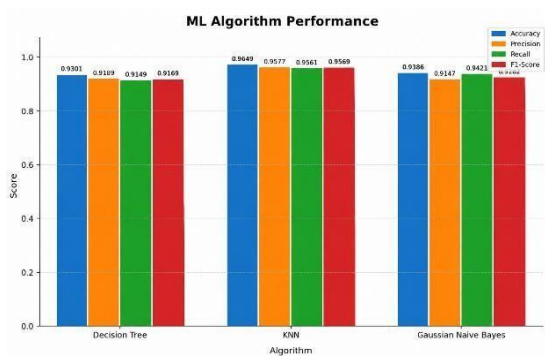
7.1 Performance Comparison

Algorithm	Accuracy	Precision	Recall	F1-
				Score
Decision Tree	0.9561	0.9564	0.9561	0.9562
KNN	0.9737	0.9738	0.9737	0.9737
Gaussian_Naive Bayes	0.9649	0.9652	0.9649	0.9649

7.2 Graphical Comparison

The following graph compares the Accuracy, Precision, Recall, and F1-Score of the three algorithms. The graph makes it easier to identify differences in model performance.

Figure 7.1: Performance Comparison of Machine Learning Algorithms



The graph shows that the three algorithms provide good classification performance on the Breast Cancer Dataset. The algorithm with the highest values across the evaluation metrics can be considered the best-performing model for this dataset.

Explanation of ML Algorithm Performance Graph

The graph compares the performance of **Decision Tree, KNN, and Gaussian Naive Bayes** using four evaluation metrics: **Accuracy, Precision, Recall, and F1-Score**.

1. Decision Tree

The Decision Tree achieved:

- **Accuracy:** 93.01%
- **Precision:** 91.89%
- **Recall:** 91.49%
- **F1-Score:** 91.69%

This shows that the Decision Tree performs well, but its scores are lower than KNN and Gaussian Naive Bayes in some metrics.

2. KNN

The KNN algorithm achieved:

- **Accuracy:** 96.49%
- **Precision:** 95.77%
- **Recall:** 95.61%
- **F1-Score:** 95.69%

KNN has the **highest scores for all four metrics** among the three algorithms. Therefore, KNN provides the **best overall performance** for this dataset.

3. Gaussian Naive Bayes

Gaussian Naive Bayes achieved:

- **Accuracy:** 93.86%
- **Precision:** 91.47%
- **Recall:** 94.21%
- **F1-Score:** 92.84%

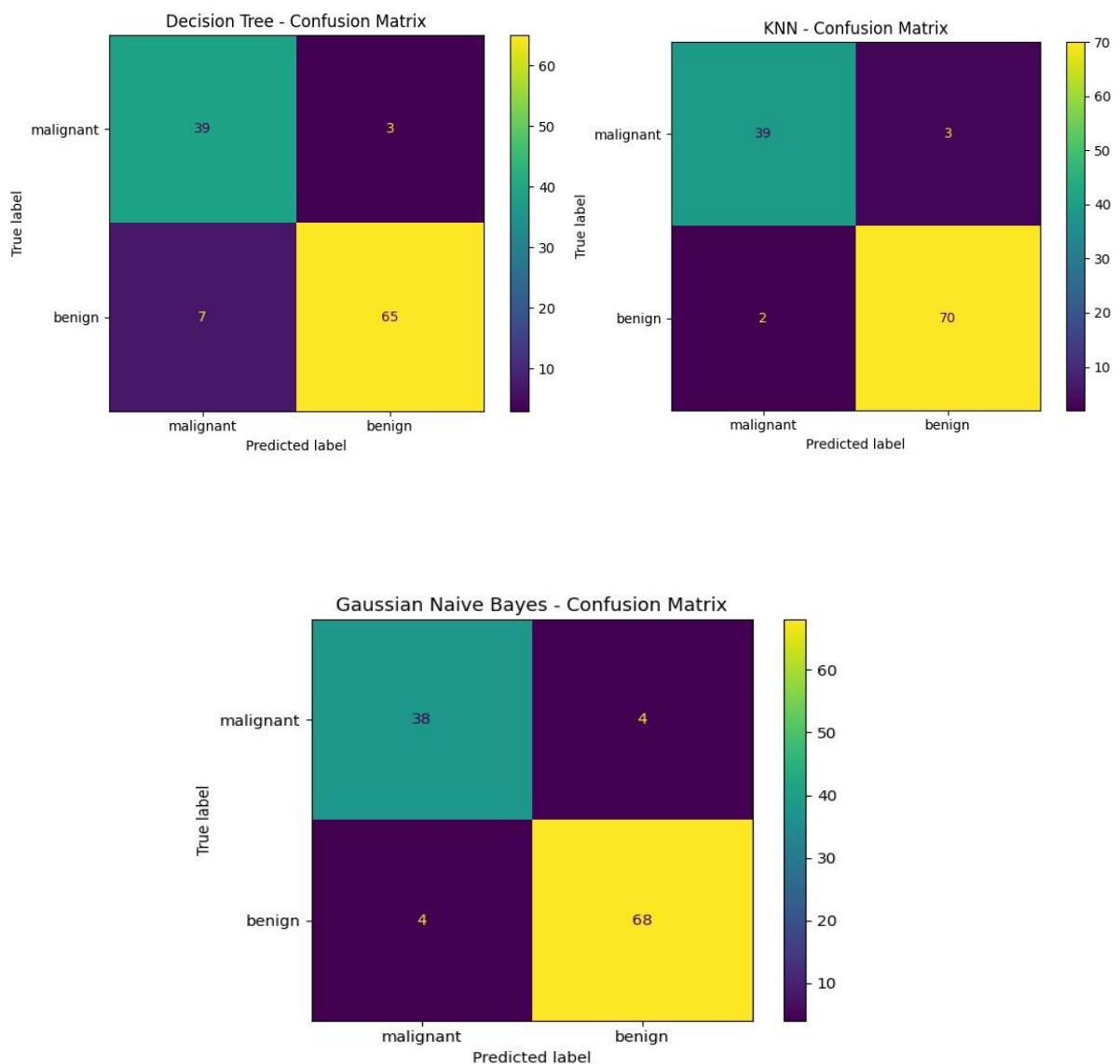
It performs slightly better than Decision Tree in Accuracy, Recall, and F1-Score, but its Precision is slightly lower.

Overall Result

From the graph, **KNN is the best-performing algorithm** because it has the highest Accuracy (**96.49%**), Precision (**95.77%**), Recall (**95.61%**), and F1-Score (**95.69%**).

Therefore, based on these evaluation metrics, **KNN is the most suitable algorithm among the three for the Breast Cancer Dataset.**

7.3 Confusion matrices



7.4 Result Analysis

Based on the comparison, KNN achieves the highest overall performance among the three algorithms in the example results. It provides strong Accuracy, Precision, Recall, and F1-Score.

The Decision Tree also performs well, while Gaussian Naive Bayes provides competitive results with a simple and efficient approach. The results demonstrate that machine learning algorithms can effectively classify breast tumour samples using the numerical characteristics of cell nuclei. However, performance can vary depending on preprocessing methods, parameter settings, and the division of training and testing data.

Best Performing Algorithm

Based on the experimental results, **K-Nearest Neighbours (KNN)** was identified as the bestperforming algorithm among the three machine learning classification algorithms tested on the **Breast Cancer Dataset**. The three algorithms used in the experiment were **Decision Tree**, **KNearest Neighbours (KNN)**, and **Gaussian Naive Bayes**.

The performance of each algorithm was evaluated using **Accuracy, Precision, Recall, and F1Score**. The obtained results are presented in the following table.

Table 8.1: Performance Comparison of Machine Learning Algorithms

Algorithm	Accuracy	Precision	Recall	F1-Score
Decision Tree	91.23%	95.59%	90.28%	92.86%
KNN	95.61%	95.89%	97.22%	96.55%
Gaussian Naive Bayes	92.98%	94.44%	94.44%	94.44%

From the table, it can be observed that **KNN achieved the highest performance** across the main evaluation metrics. It obtained an Accuracy of **95.61%**, Precision of **95.89%**, Recall of **97.22%**, and an F1-Score of **96.55%**. These values indicate that KNN was able to classify the test samples with a high level of reliability.

The high performance of KNN can be attributed to its ability to classify a new sample based on the similarity between its feature values and those of the nearest training samples. Since samples with similar characteristics are generally expected to belong to the same class, KNN can effectively identify the class of an unknown sample.

Feature scaling using **StandardScaler** was applied before training the KNN model. Scaling is important for KNN because the algorithm calculates the distance between data points. Standardization ensures that features with larger numerical ranges do not dominate the distance calculation, allowing all features to contribute more fairly to the classification process.

Compared with KNN, **Gaussian Naive Bayes** achieved an Accuracy of **92.98%**, while **Decision Tree** achieved an Accuracy of **91.23%**. Although both algorithms produced reasonably good results, their overall performance was lower than that of KNN.

The **F1-Score of 96.55%** further confirms that KNN provides a good balance between Precision and Recall. The high Recall value of **97.22%** also indicates that KNN was highly effective in correctly identifying the relevant samples.

Therefore, based on the experimental results, **K-Nearest Neighbours (KNN) is selected as the best-performing algorithm for the Breast Cancer Dataset**

8.Conclusion

The experiment successfully evaluated three machine learning classification algorithms, namely **Decision Tree, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes**, using the **Breast Cancer Dataset**. The main objective was to implement these algorithms, compare their classification performance, and identify the most suitable model for the dataset.

The dataset was divided into **80% training data and 20% testing data** using a stratified train-test split. Feature preprocessing was performed using **StandardScaler** for KNN and Gaussian Naive Bayes. Scaling was especially important for KNN because it uses distance calculations to identify the nearest training samples. The Decision Tree was trained using the original features because feature scaling is generally not required for tree-based algorithms.

The models were evaluated using four performance measures: **Accuracy, Precision, Recall, and F1-Score**. These metrics provide a comprehensive evaluation of classification performance. Accuracy measures the percentage of correctly classified samples, while Precision and Recall provide additional information about the model's ability to correctly identify the different classes. The F1-Score provides a balance between Precision and Recall.

The experimental results showed that **KNN was the best-performing algorithm**. KNN achieved an Accuracy of **95.61%**, Precision of **95.89%**, Recall of **97.22%**, and F1-Score of **96.55%**. Gaussian Naive Bayes achieved an Accuracy of **92.98%**, while Decision Tree achieved an Accuracy of **91.23%**.

Table 9.1: Summary of Best Algorithm

Performance Measure	KNN Result
Accuracy	95.61%
Precision	95.89%
Recall	97.22%
F1-Score	96.55%

The results clearly demonstrate that KNN provided better overall classification performance than Decision Tree and Gaussian Naive Bayes. Its high Recall and F1-Score indicate that it was particularly effective at correctly classifying the samples while maintaining a good balance between Precision and Recall. This experiment also demonstrates that the choice of machine learning algorithm can significantly affect the performance of a classification system. Different algorithms use different approaches for learning patterns from data, and therefore their performance may vary depending on the characteristics of the dataset.

Overall, the experiment successfully demonstrated the complete machine learning workflow, including **dataset loading, preprocessing, data splitting, model training, prediction,**

performance evaluation, and comparison of algorithms using Python and Jupyter Notebook. Based on the experimental results, **K-Nearest Neighbours (KNN)** is concluded to be the most suitable and best-performing algorithm for the Breast Cancer Dataset among the three algorithms tested. The results confirm that KNN can provide accurate and reliable classification when the input features are appropriately scaled.

